

Toni Ihab Youssef *Artificial Intelligence Senior*

tony0100512@cs.bnu.edu.eg | [linkedin.com/in/toni-ihab](https://www.linkedin.com/in/toni-ihab) | github.com/tokihab | <https://huggingface.co/TOKII> | 01224477633

Roxy, Heliopolis | Fluent in English, German B1 and an Arabic native

Artificial Intelligence Engineer with proven expertise in NLP pretraining, real-time voice/LLM pipelines, and medical imaging. Strong foundation in cloud infrastructure, DevOps automation, and testing frameworks. Seeking roles in Applied ML or AI/MLOps where independent technical autonomy and architectural excellence are valued.

EDUCATION

English School / El-Nasr School, ESC 2008 – 2022
Graduated 75% | RoboCup Participant during Secondary School | **2nd Place** District Coding Competition during High School
Heliopolis, Cairo

Benha National University, Bachelor of Computer Science 2022 – 2026
GPA: 3.02 | GDG Cybersecurity Learner | ML Core **Technical Advisor** | ICPC Competitor
Obour, Cairo

EXPERIENCE

Digital Egyptian Pioneer Initiative, Freelance Trainee 10/2024 – 06/2026
- Applied business development and technical project management under mentor guidance
- **Delivered 4+** independent freelance projects including UI/UX, academic diagrams, translation and data annotation tasks
Online

National Media Authority, Infrastructure Security Fellow 06/2024 – 07/2024
- **Evaluated** physical and logical security of national-scale server environments and broadcast systems
- Completed seminars on network defense, incident response, and mission critical IP infrastructure
Maspero

Arabic African International Bank, Operations/Data Intern 08/2024 – 09/2024
- Processed end-to-end checks workflows, ensuring contract compliance and secure data entry
- Engineered data pipelines and **PowerBI dashboards** for operational insights
Digital Factory, New Cairo

COURSES

DevOps, Digital Egyptian Pioneer Initiative 12/2025 – 07/2026
- Automated infrastructure provisioning and containerized **E-commerce migration**, improving deployment speed
- Accomplished this after learning the following: Ansible, Terraform, AWS, Docker, Kubernetes, Jenkins, CI/CD, Prometheus and Grafana.
YAT, Ard El-Golf

Cloud Architecture, AWS 03/2025 – 04/2025
- Compute, Storage, Databases, Networking, Security, Auto Scaling, Load Balancing
- **Applied these technologies** on a Coffee Shop's web app inside AWS sandbox during the courses period
National Telecom Institute

Software Testing, Digital Egyptian Pioneer Initiative 05/2025 – 11/2025
- **Applied ISTQB methodologies** we learnt to manage the full testing lifecycle of an **E-commerce platform**
- Engineered testing suites transitioning from manual A/B testing exploratory sessions to Data-Driven and REST API testing, utilizing Selenium and Java's TestNG to validate complex back-end service layers and data integrity.
WE, Nasr City

CERTIFICATES

Data Engineering, Coursera/IBM
- Learned advanced level workflows for excel and created **time-series forecasts**.
- Analyzed large datasets and performed **EDA, statistical analysis and feature engineering pipelines** that go hand in hand with the models i built.
- Applied these skills in building predictive models and Business Dashboards about **sustainable fashion, supply chains and robots kinematics**.

Fullstack Web Development, Coursera/University of Michigan
- Created static web clones of youtube and facebook with just HTML and CSS
- Learned about PDO PHP connections to SQL DBs
- Deep dive into dotnet with its MVC structure and how to visualize it using react framework's components
- Applied concepts in personal project: **Tier List Website** (React/.NET/PHP/MySQL)

CCNA Routing and Switching, CISCO/part of my Bachelor's curriculum
- Mastered IP addressing, subnetting, and switching concepts
- Implemented network configurations using **Cisco Packet Tracer**

PROJECTS

MASRISENSE, PyTorch, Hugging Face, Transformers
- Curated MASRISET (4.5M+ Egyptian Arabic rows) and continued MLM pretraining on MASRIBERTv3 (**perplexity: 16**).
- Evaluated **emotion/sarcasm/sentiment** multi-task detection using F1-score and confusion-matrix analysis.

- Built a multi-scale acoustic model (CNN + BiLSTM + Transformer) to **extract prosody and emotion features** from 16 kHz audio.
- Implemented a late-fusion engine with **8 decision rules** to resolve conflicts between sarcasm and text sentiment signals.
- Context-tracking layer monitoring **5-turn conversation** trends for dynamic risk assessment and de-escalation recommendations

Kalamna's Voice, *FastAPI, Pipecat, LiveKit, WebSockets*

- Ultra-low-latency WebRTC using **Livekit's infrastructure**, STT → LLM → TTS pipeline with **Pipecat's orchestration**.
- Optimized inference latency for real-time conversations by using asynchronous frame through VAD handling and non-blocking I/O.
- Implemented structured **logging** for turn events and conversation state to support **monitoring and debugging**.
- **Deployed** the pipeline as a model-serving backend with FastAPI inference endpoints.

Medical X-Ray Caption Generator, *TensorFlow, Keras, Attention*

- Dual-model system comparing **custom 4-block** CNN encoder vs. ResNet for medical X-ray analysis
- Bahdanau **Attention mechanism** computing alignment scores; GRU/LSTM decoder focuses on specific image regions
- **Deployed working prototype** on Hugging Face Spaces for inference

Movie Recommender, *NLP, Machine Learning, Py*

- Built collaborative filtering and content-based recommendation system for **personalized movie suggestions**
- **Deployed end-to-end pipeline** with model evaluation and user-facing recommendations

3-DOF RPR Robotic Manipulator Simulation, *MATLAB with Peter Corke's Robotics Toolbox*

- SerialLink model with Denavit-Hartenberg parameters; **forward and inverse kinematics** with joint limits
- Smooth joint-space trajectories via jtraj; validated reachability and **captured automated motion data for further analysis**

Image Processing Toolkit, *MATLAB*

- **30+ functions** for spatial/frequency domain operations: median, Gaussian, Butterworth filtering, edge detection, histograms
- **GUI application** enabling real-time image manipulation with adjustable parameters and live visualization

Linear Algebra Calculator, *C++*

- Object-oriented matrix/vector calculator with **10+ operations**; Gauss-Jordan elimination with partial pivoting for $n \times n$ systems

Pong Game, *Assembly x8086*

- **Game logic and GUI implemented** in low-level assembly; DOS simulation

SKILLS

NLP, LLMS:

Transformers, MLM pretraining, sentiment/emotion/sarcasm classification

Speech/Voice:

STT, TTS, WebRTC, VAD, real-time pipelines

ML frameworks:

PyTorch, TensorFlow/Keras, Langchain

MLOps/Cloud

FastAPI, Docker, Kubernetes, AWS, Terraform, Ansible, CI/CD, Jenkins, AWS

Testing

Selenium, TestNG, REST API testing (ISTQB)

Other

Prolog, Verilog, CLIPS, LaTeX, ModelSIM, Proteus, Spyder, Eclipse, MobaXterm, Tableau

VOLUNTEERING

Machine Learning Technical, *Teaching/Guiding freshmen year students basic of AI*

01/2024 – 04/2024
GDG on BNU Campus

Cyber Security Member, *Learning from senior year students basics of Security*

09/2023 – 12/2023
GDG on BNU Campus